Scheme For Valuation
MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026
FOURTH SEMESTER (2024 ADMISSION ONWARDS)

MG4DSCSTA200 – BASICS OF MULTIVARIATE DISTRIBUTIONS

Duration: 1Hour 30 min Maximum marks: 50

Part A
Answer any seven questions. Each question carries 2 marks.

1. For explaining why two regression lines arise. (2 Marks)
2. For describing Kendall's Tau coefficient. (2 Marks)
3. Proof (2 Marks)
4. Statement (1 Mark)
   Proof (1 Mark)
5. Any two properties of multivariate normal distribution. (2 Marks)
6. Any two examples which follows multinomial distribution or pmf of multinomial distribution. (2 Marks)
7. Pdf of MVN (1 Mark)
   Likelihood function of MVN (1 Mark)
8. Mean=$l'\mu$, Variance= $l'\Sigma\, l$ (2 Marks)
9. $X^{(1)}$ follows $N(\mu^{(1)},\Sigma_{11})$ (2 Marks)
10. $Var(X'AX)= 2tr(A\Sigma A\Sigma) +4\mu'A\Sigma A\mu$ (2 Marks)

Part B
Answer any 4 questions. Each question carries 6 marks.

11. Statement (1 Mark)
    Proof (5 Marks)
12. Statement Spearman's rank correlation coefficient formula. (1 Mark)
    Proof. (5 Marks)
13. (1) For setting U=X+Y and $R_U = \{1,2,3,4,5\}$ (1 Mark)

u	1	2	3	4	5
g(u)	3/72	11/72	24/72	21/72	13/72

(2 Marks)

(2) $f_1(x|X+Y=3)=\frac{9-x}{24},\ \ x=0,1,2$ (2 Marks)

(3) X and Y are not independent. (1 Mark)

14. $k=\frac{4}{35}$ (1 Mark)

$f_1(x)=\frac{2}{7}(x+3);\ \ 0<x<1$ (2 Mark)

$f_2(y)=\frac{2}{5}(y+2);\quad 0<y<1$ (2 Mark)

X and Y are independent (1 Mark)

15. Derivation of necessary and sufficient condition for independence of two sub vectors of a MVN vector. (6 Mark)

16. Proof of independence of X'AX and X'BX when AΣB=0 (6 Mark)

Part c

Answer any 1 questions. Each question carries 12 marks.

17. (a) Description of Scatter Diagram (3 Marks)

18. Interpreting the Regression (3 Marks)

19. Interpreting Correlation (3 Marks)

(b) For constructing scatter diagram. (3 Marks)

20. Proof of X'AX follows chi square with r df. (9 Mark)

Proof of A is idempotent with rank r. (3 Mark)